

The following arguments are supported by the *plot* function:

Argument	Description
x	A vector for the x-axis
y	The vector or list in the y-axis
type	'p' for points, 'l' for lines, 'o' for overplotted plots and lines, 's' for stair steps, 'h' for histogram
xlim	The x limits of the plot
ylim	The y limits of the plot
main	The title of the plot
sub	The subtitle of the plot
xlab	The label for the x-axis
ylab	The label for the y-axis
axes	Logical value to draw the axes

The line chart is shown in Figure 1.

The autocorrelation plot can be used to obtain correlation statistics for time series analysis, and the same can be generated using the *acf* function in R. You can specify the following autocorrelation types: *correlation*, *covariance*, or *partial*. Figure 2 shows the ACF chart that represents the CPI values ('x' in the chart) for the state of Punjab.

The function *acf* accepts the following arguments:

Argument	Description
x	A univariate or multivariate object or vector or matrix
lag.max	The maximum lag to calculate the acf
type	Supported values 'correlation', 'covariance', 'partial'
plot	The acf is plotted if this value is TRUE
i	A set of time difference lags to retain
j	A collection of names or numbers to retain

Bar chart

The *barplot* function is used to draw a bar chart. The chart for Punjab's CPI can be generated as follows, and is shown in Figure 3:

```
> barplot(punjab$x, main="Punjab Consumer Price Index",
sub="Upto November 2021", xlab="Year", ylab="Consumer Price
Index", col="navy")
```

The function is quite flexible and supports the following arguments:

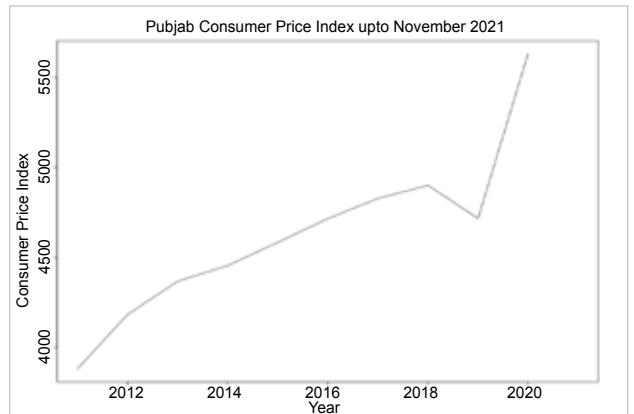


Figure 1: Line chart

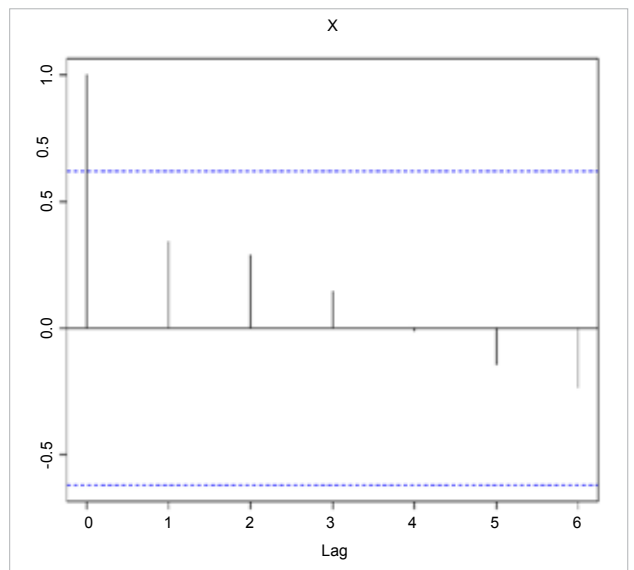


Figure 2: ACF chart

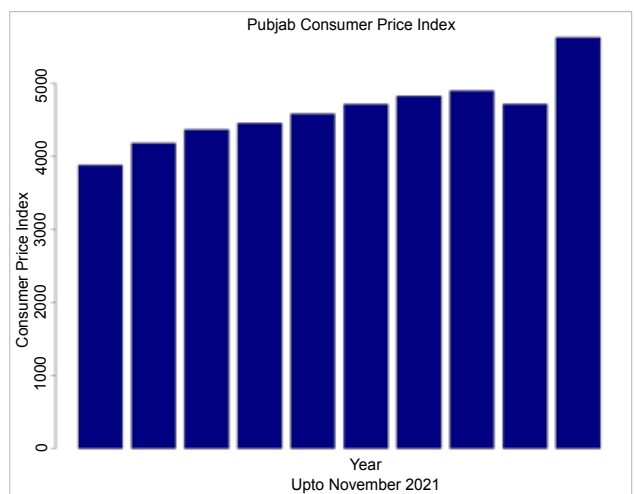


Figure 3: Line chart of Punjab's CPI